L-IDS: Project Defense Visual Guide

This guide uses analogies and visuals to help you explain the core technical concepts of your project to your professor.

1. The "Guard Outside the Window" Analogy

Use this to explain why Hypervisor-level monitoring is superior to traditional antivirus.

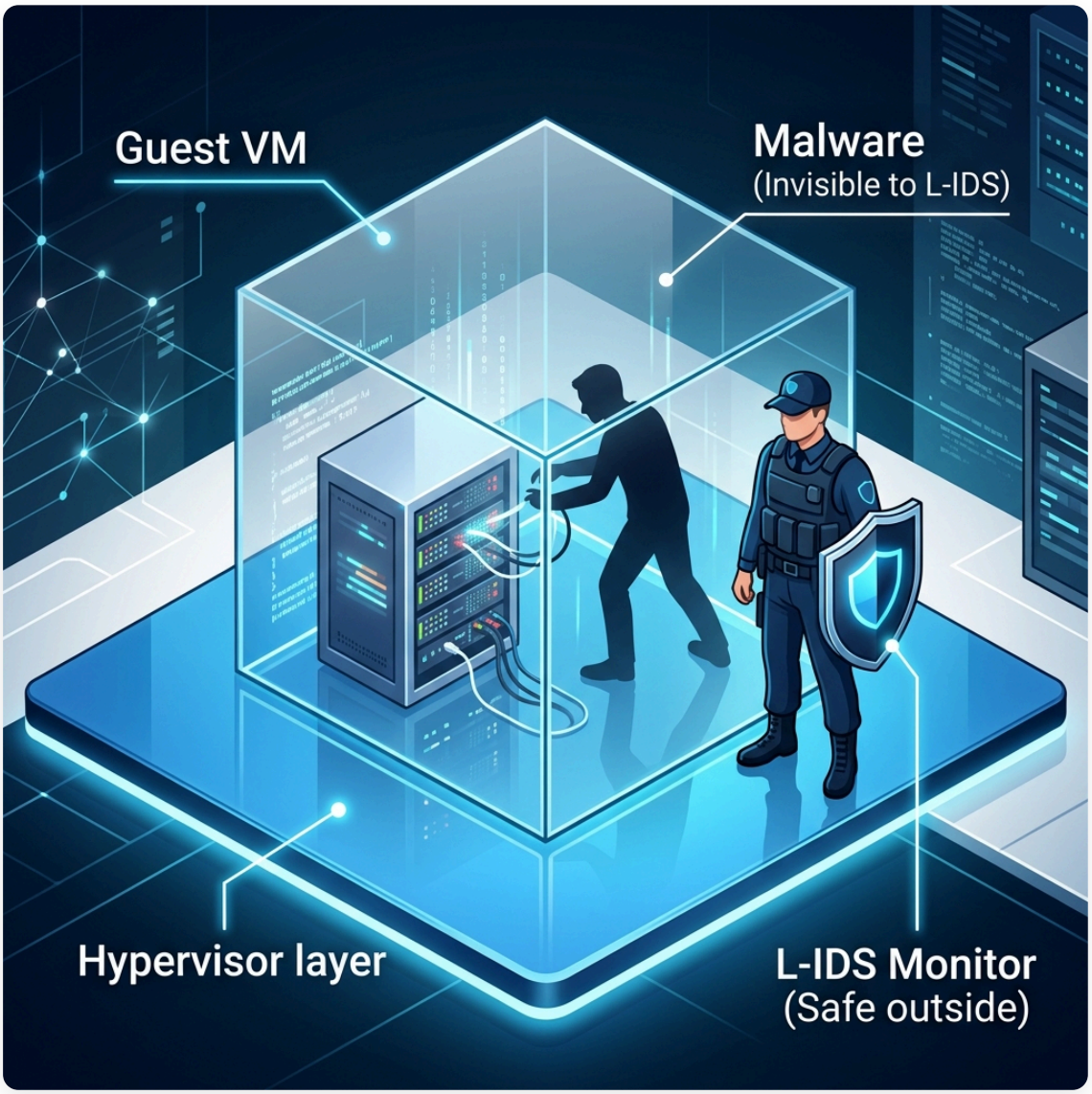


Figure 1: Comparison between Guest-level and Hypervisor-level monitoring

> **Presentation Script:** "Professor, imagine a Cloud VM is like a room with a burglar inside. Traditional antivirus is like a guard inside the room—malware can hide from or attack it. Our L-IDS is like a guard outside the window (the Hypervisor). The burglar can't reach the guard, but the guard sees everything."

2. Why Lightweight Statistical Detection?

Use this to explain why you chose statistical methods over heavy AI/Machine Learning.

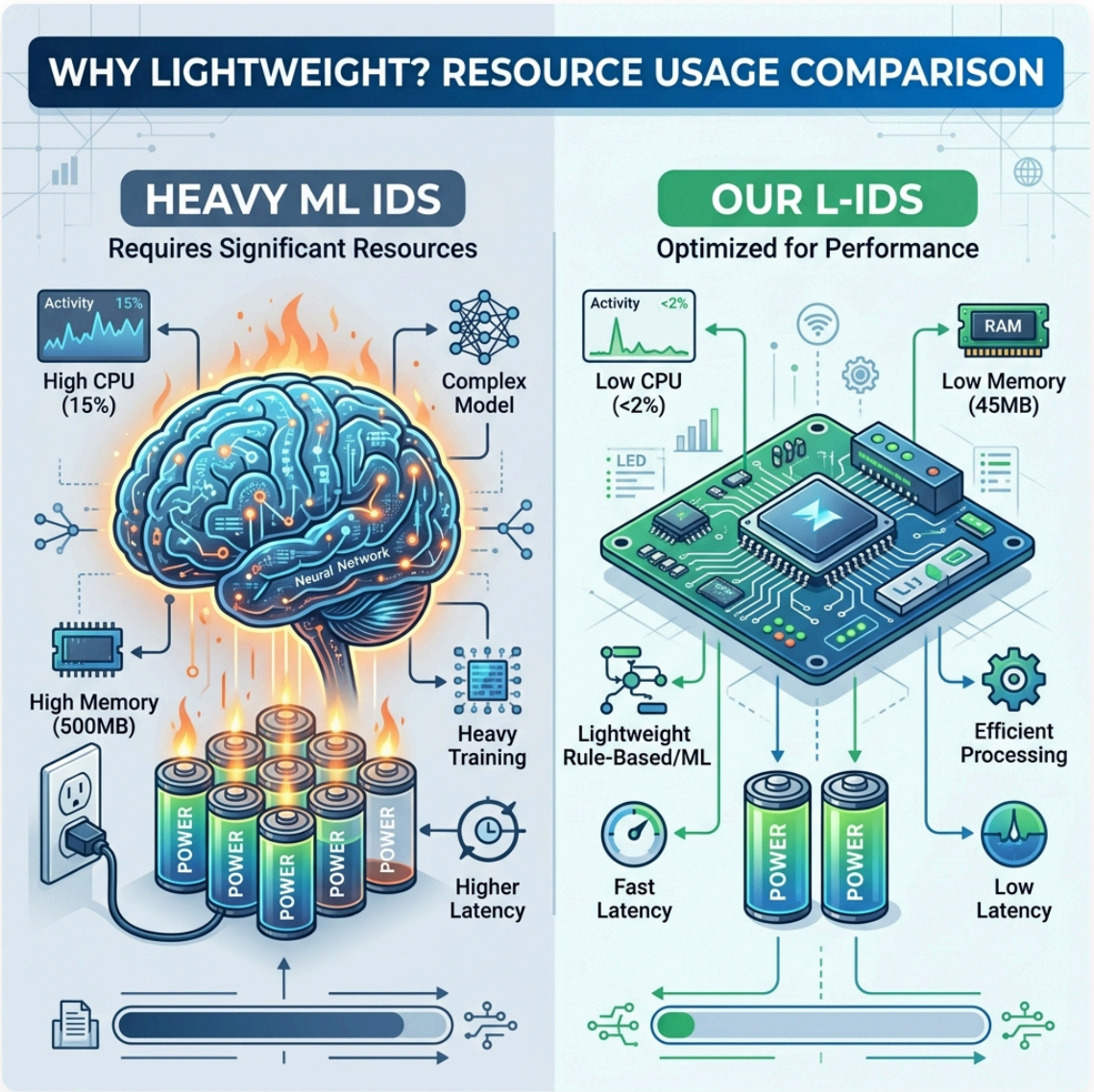
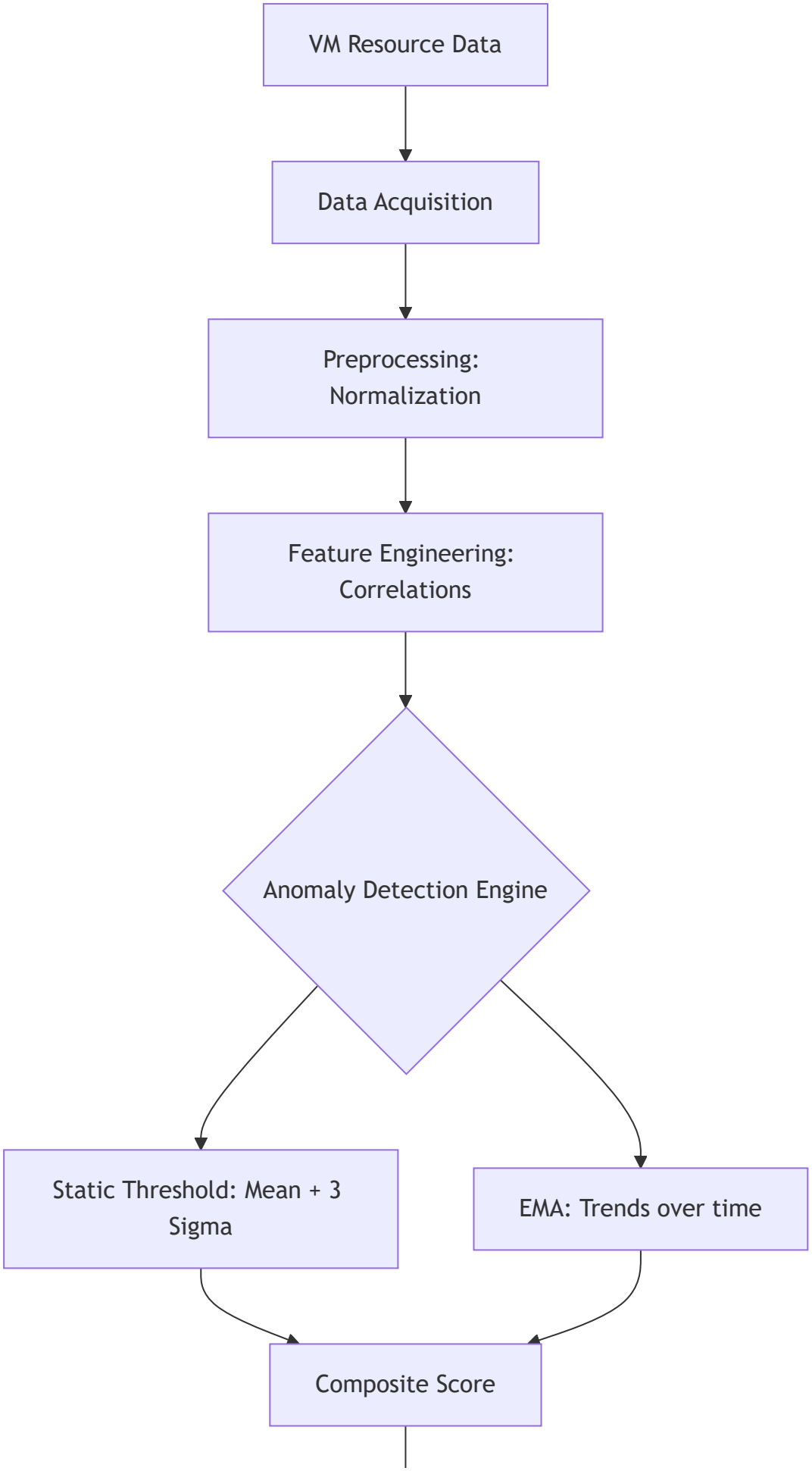


Figure 2: Resource overhead comparison: Statistical vs. ML approaches

> **Presentation Script:** "Cloud resources are expensive. Most AI/ML systems use 10-15% of the server's CPU just for security. Our system uses basic statistics to achieve 92% accuracy with less than 2% CPU overhead, making it practical for real-world business use."

3. The Detection Pipeline

Use this to show the mathematical logic of your system.


VM Resource Data
Data Acquisition
Preprocessing: Normalization
Feature Engineering: Correlations
Anomaly Detection Engine
Static Threshold: Mean + 3 Sigma
EMA: Trends over time
Composite Score

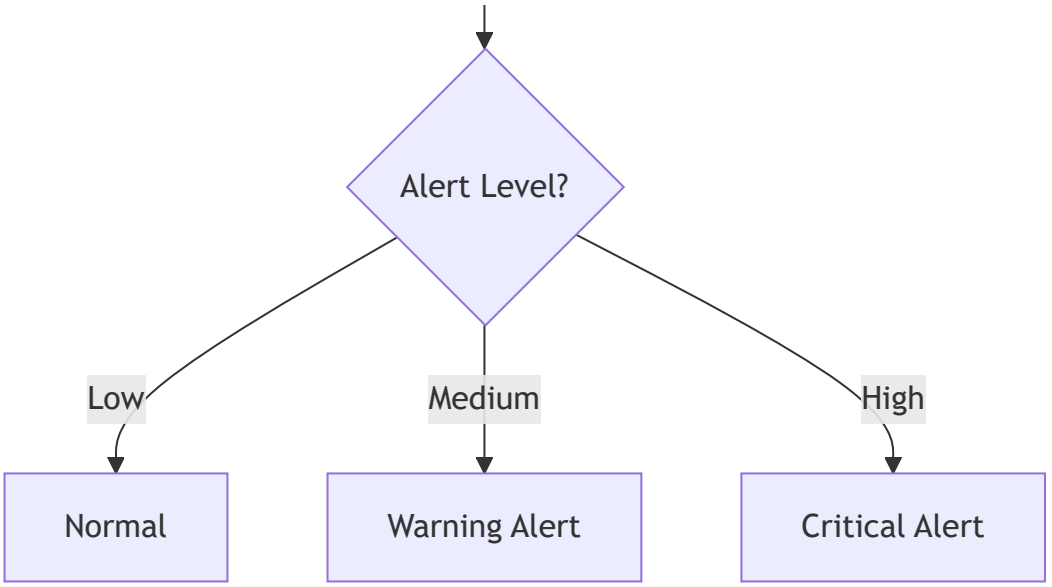


Figure 3: 6-stage data processing and detection pipeline